

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2025

Programme: BE

Full Marks : 100

Course: Digital Signal Analysis and Processing (New)

Pass Marks : 45

Time : 3 hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

1. a) Define Signal. Explain the basic elements of DSP with block diagram. List the advantages of DSP over analog signal processing. 7

OR

Differentiate between energy and power signal. Determine whether the following systems are linear or not.

i.  $y(n) = ax(n) + 6$

ii.  $y(n) = x(n) + nx(n+1)$

- b) What is the input signal  $x(n)$  that will generate the output sequence  $y(n) = [8, 22, 11, 31, 4, 12]$  for a system with impulse response  $h(n) = [2, 5, 0, 4]$ ? 8
2. a) Define Z-transform ROC of a signal  $x[n]$ . Determine the z-transform and ROC of  $x(n) = [5(2^n) - 9(3^n)] u(n)$ . 8
- b) Determine the inverse Z - transform 7

$$X(Z) = \frac{1}{1 + \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

When ROC:  $|Z| > 1$  and ROC:  $|Z| < \frac{1}{2}$

3. a) Obtain direct form I and direct form II for the system described by  $y(n) = -0.1 y(n-1) + 0.72 y(n-2) + 0.7 x(n) - 0.252 x(n-2)$ . 8
- b) Draw the lattice structure for the following FIR filter function  $H(z) = 1 + 2z^{-1} + \frac{1}{3}z^{-2}$ . 7
4. a) Derive Impulse Invariant Method for IIR filter design. Also illustrate the mapping from s-plane to the z-plane while using IIM. 7

- b) Using BLT design a Butterworth filter which satisfies the following conditions: 8

$$\begin{aligned} 0.8 \leq |He^{jw}| \leq 1 & \quad 0 \leq |w| \leq 0.2\pi \\ |He^{jw}| \leq 0.2 & \quad 0.6\pi \leq |w| \leq \pi \end{aligned}$$

5. a) Explain the design procedure for low pass digital Butterworth IIR filter using bilinear transformation method. 8

Assume

$A_1$  = Gain at a passband frequency  $\omega_1$

$A_2$  = Gain at a stopband frequency  $\omega_2$

$\omega_1$  = Analog frequency corresponding to  $\omega_1$

$\omega_2$  = Analog frequency corresponding to  $\omega_2$

OR

Draw the flowchart of Remez exchange algorithm and explain it.

- b) Explain the FIR Filter designing steps using Kaiser Window. 7
6. a) Find the 8-point DITFFT of  $x(n) = \sin \frac{\pi n}{3}$  8
- b) Prove that multiplication of two sequences gives the result in circular convolution of these two sequences. 7
7. Write short notes on: (Any two) 2×5
- a) DTFT and properties
- b) LTI system
- c) Linear constant co-efficient difference equation